

Rectangular block-encoding: SVD vs dilation

Shapes (M_rows x M_cols): [(4, 208), (22, 208), (208, 4), (208, 22)].
K values: [1, 8, 27, 64, 125, 216], phase_bitsize = 32.

Strategy (rank-aware):

SVD: pad to $N \times N$ with $N = \text{next_pow2}(\max(M_rows, M_cols)) = 256$.

Use rank $r = \min(M_rows, M_cols)$ so the U -, V -isometries need only r partial-interferometer layers, and the diagonal Sigma loads r values per block. Cost = $2 \times \text{interferometer}(K, N, n_layers=r) + \text{diag}((K, r))$.

Toffoli scales linearly in r : $r=4$ vs $r=22$ gives a $\sim 5\times$ gap at fixed K .

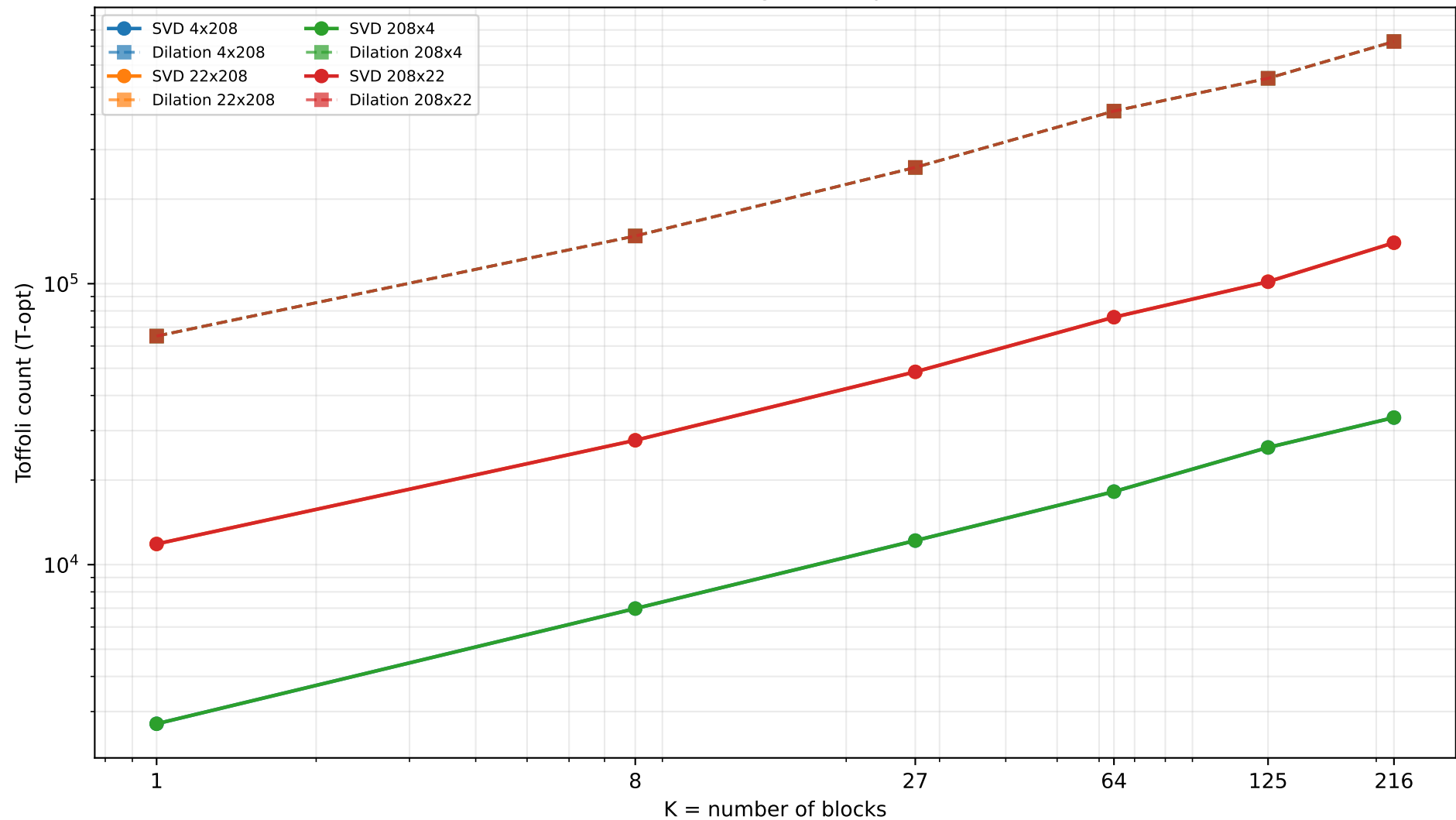
Dilation: Stinespring W is a unitary of size $(M_rows + M_cols)$. Pad to

$N_dil = \text{next_pow2}(M_rows + M_cols)$. Cost = $\text{interferometer}(K, N_dil)$.

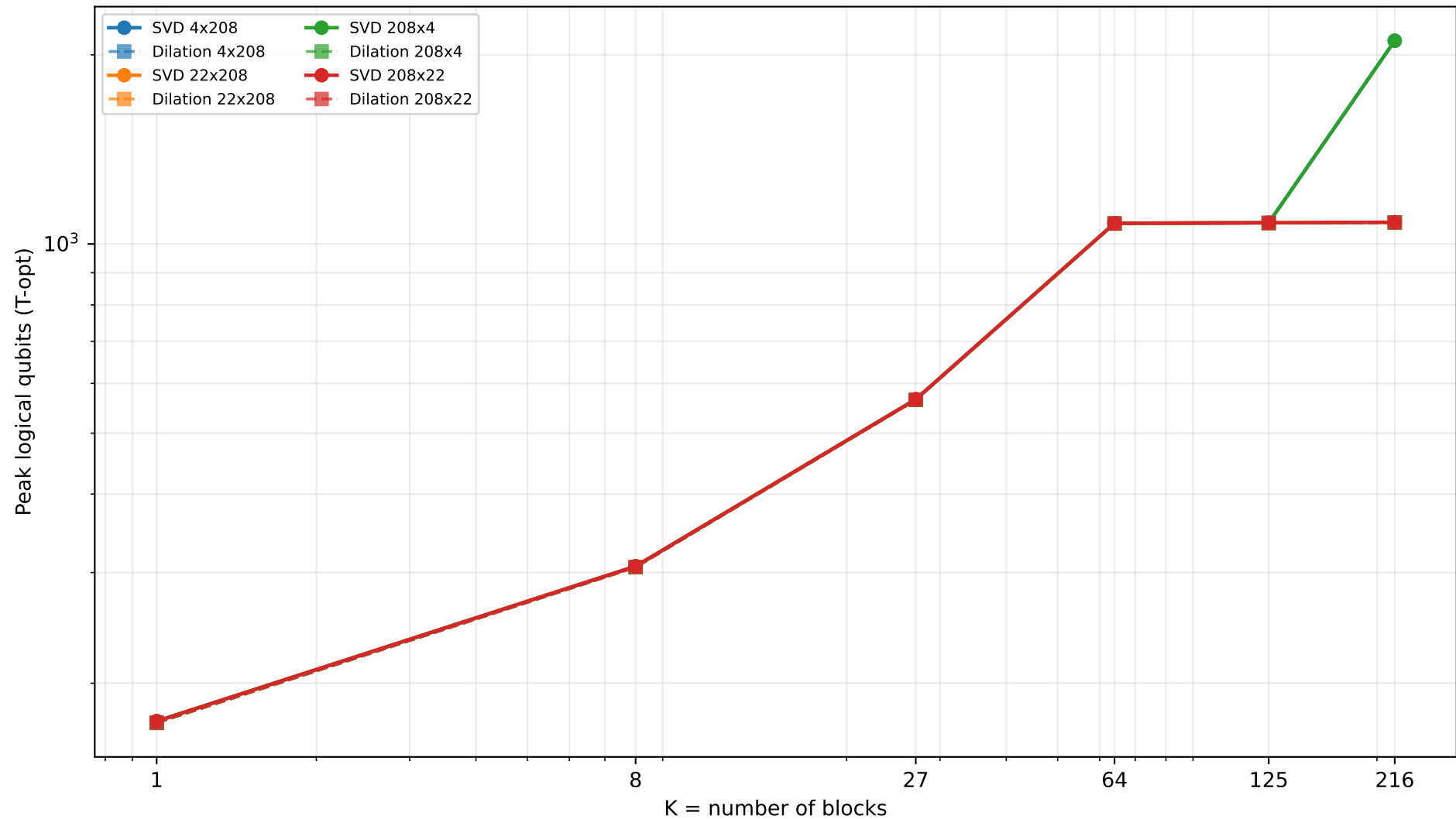
Independent of which side is the small dimension.

All Toffoli/qubit numbers come from Qualtran's `get_cost_value(QECGatesCost)`
walking each bloq's `build_call_graph` - no analytic formulas.

Toffoli vs K, rectangular shapes (b=32)



Peak qubits vs K (b=32)



Rectangular shape resource counts (b=32)

shape	rank r	K	N_SVD	SVD T	SVD Q	N_dil	Dil T	Dil Q	SVD/Dil
4 x 208	4	1	256	2,714	174	256	65,082	173	0.042
4 x 208	4	8	256	6,978	307	256	147,836	306	0.047
4 x 208	4	27	256	12,179	566	256	259,264	565	0.047
4 x 208	4	64	256	18,194	1079	256	411,400	1078	0.044
4 x 208	4	125	256	26,124	1081	256	538,280	1080	0.049
4 x 208	4	216	256	33,378	2106	256	727,560	1082	0.046
22 x 208	22	1	256	11,846	174	256	65,082	173	0.182
22 x 208	22	8	256	27,709	307	256	147,836	306	0.187
22 x 208	22	27	256	48,554	566	256	259,264	565	0.187
22 x 208	22	64	256	75,978	1079	256	411,400	1078	0.185
22 x 208	22	125	256	101,621	1081	256	538,280	1080	0.189
22 x 208	22	216	256	139,875	1083	256	727,560	1082	0.192
208 x 4	4	1	256	2,714	174	256	65,082	173	0.042
208 x 4	4	8	256	6,978	307	256	147,836	306	0.047
208 x 4	4	27	256	12,179	566	256	259,264	565	0.047
208 x 4	4	64	256	18,194	1079	256	411,400	1078	0.044
208 x 4	4	125	256	26,124	1081	256	538,280	1080	0.049
208 x 4	4	216	256	33,378	2106	256	727,560	1082	0.046
208 x 22	22	1	256	11,846	174	256	65,082	173	0.182
208 x 22	22	8	256	27,709	307	256	147,836	306	0.187
208 x 22	22	27	256	48,554	566	256	259,264	565	0.187
208 x 22	22	64	256	75,978	1079	256	411,400	1078	0.185
208 x 22	22	125	256	101,621	1081	256	538,280	1080	0.189
208 x 22	22	216	256	139,875	1083	256	727,560	1082	0.192